

Statistical outputs

```
#-----  
#import libraries  
library(ggspatial)  
library(cowplot)  
library(stringr)  
library(dplyr)  
library(raster)  
library(sf)  
#library(maptools)  
library(stars)  
library(lubridate)  
library(data.table)  
library(ggplot2)  
library(purrr)  
library(lme4)  
library(lmerTest)  
library(effects)  
library(introdataviz) # split violin plots  
library(extrafont)
```

Prepare dataset

```
#-----  
options(scipen=999)  
Sys.setenv(TZ='UTC') # change time zone for session  
#setwd("D:/OneDrive - NIOZ/")  
#setwd("C:/Users/cbeardsworth/OneDrive - NIOZ")  
setwd("C:/Users/BESCBEAR/OneDrive - NIOZ")  
  
ele <- raster("2_Reference Data/Elevation/griend_bathy_2015-2020.tif")  
SA <- read.csv("2_Reference Data/Water/spacebywaterlevel.csv")  
griend_poly <- read_sf("2_Reference Data/Griend/Griend_2015-2020Bathy_100NAP.gpkg")  
meta <- read.csv("2_Reference Data/Bird Metadata/2018-2022_islandica.csv") %>%  
  dplyr::select(release_ts, tag, captivity, age)  
  
waterdn <- read.csv("2_Reference Data/Water/2017-2022_waterlevel_under50_propdaynight.csv") %>% # propo  
  mutate(total_light = prop_day + prop_civil,  
         total_dark = prop_naut + prop_astro + prop_night,  
         tod_summary = case_when(total_light >= 0.70 ~ "day", #tide majority in day  
                                total_light <0.70 & total_dark <0.70 ~ "twilight",  
                                total_dark >=0.70 ~ "night")) #tide majority at night  
  
tides <- read.csv("2_Reference Data/Water/Griend/Wide/HLH/allYears-tidalPattern-HLH-griend_Wide-UTC.csv")
```

```

mutate(year = substring(tideID,1,4),
       yearly_id = as.numeric(substring(tideID, 6,8)),
       tideID = as.numeric(sub("_", "", tideID)),
       high_start_time = as.POSIXct(high_start_time),
       date = as.Date(high_start_time, tz="UTC"),
       month = month(high_start_time)) %>%
left_join(SA, by = c("low_level"="waterlevel")) %>% #space available in the tide
left_join(waterdn, by="tideID") %>%
filter(month %in% c(9,10), tod_summary!="twilight" & start_tod == tod_summary) %>% #choose only aft
rename(tide_tod = start_tod)

#choose residence patches that are not roosting (<50NAP), adults only, not our birds that were in capti
df <- fread("1_WATLAS/Data/res_patches/2018-2022_patches.csv") %>%
as.data.frame() %>%
filter(!tag %in% c(911,923), captivity == "no", age==3) %>% #choose birds for analysis
mutate(tod_start = ifelse(tod_start %in% c("day", "civil twilight"), "day", "night"),
       datetime_start_UTC = as.POSIXct(datetime_start_UTC, tz = "UTC"),
       month_num = str_pad(lubridate::month(datetime_start_UTC),2,pad="0"),
       month_name = lubridate::month(datetime_start_UTC, label=T),
       month = paste(month_num, month_name, sep="_"),
       tag = str_pad(tag, 4, pad = "0"),
       bird_elevation_start = raster::extract(ele, cbind(x_start, y_start)), # get elevation for wh
       roost_start = ifelse(bird_elevation_start>=50, 1, 0),
       bird_elevation_end = raster::extract(ele, cbind(x_end, y_end)),
       roost_end = ifelse(bird_elevation_end>=50, 1, 0),
       roost_site = ifelse(roost_start ==1 | roost_end == 1, 1,0)) %>%
filter(roost_site == 0,
       tideID %in% tides$tideID) %>% # only choose tides that started at the same time of day as t
left_join(., tides[,c("tideID", "tide_tod")], by = c("tideID"= "tideID")) %>%
st_as_sf(coords = c("x_start","y_start"), crs = 32631, remove = F) %>%
mutate(dist_from_griendpoly = as.numeric(st_distance(., griendpoly)))

df$geometry <- NULL

# Calculate duration tracked per ind, per tide.

bird_tides <- df %>%
  group_by(year, tideID, tag, tide_tod) %>%
  summarise(sum_duration_min = sum(duration_s)/60, ##low tide durations, we calculate the sum of the
            furthest_patch = max(dist_from_griendpoly)) %>% ##calculate furthest distance from griend
  filter(sum_duration_min >=240) %>% # which birds on which tides should be analysed.
  left_join(tides[,c("tideID","low_level")], by = "tideID") #add low level for analysis on distance f

## 'summarise()' has grouped output by 'year', 'tideID', 'tag'. You can override
## using the '.groups' argument.

min_tides <- 5 ## choose min tides
tides_table <- as.data.frame.matrix(table(bird_tides$tag,bird_tides$tide_tod)) %>% ##which birds have a
  filter(day>= min_tides, night >= min_tides)

bird_tides <- bird_tides %>%
  filter(tag %in% row.names(tides_table)) %>% ##choose only birds that had minimum tides tracked for
  mutate(year = as.factor(year),

```

```

tag = as.factor(tag))

# sample sizes

table(bird_tides$year, bird_tides$tide_tod) ##total tides tracked in day/night

##
##      day night
## 2018 386   262
## 2019 365   195
## 2020  49    65
## 2021 410   186
## 2022 239   124

tide_sample_size <- unique(as.data.frame(bird_tides[, c("tag", "tideID", "tide_tod")))) ##number of tides
tide_sample_size <- as.data.frame.matrix(table(tide_sample_size$tag, tide_sample_size$tide_tod))
tide_sample_size$dif <- tide_sample_size$day - tide_sample_size$night
summary(tide_sample_size$day) ## how many day tides were birds tracked for

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      5.00   9.00   13.00   13.42   17.00   27.00

summary(tide_sample_size$night) ## how many night tides were they tracked for

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      5.000   6.000   7.000   7.704   9.000   16.000

summary(tide_sample_size$dif) ##what were the differences within individuals for tracking day or night

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     -7.000   2.750   5.000   5.713   9.000   17.000

sample_size <- unique(as.data.frame(bird_tides[, c("year", "tag"))))
table(sample_size$year)

##
## 2018 2019 2020 2021 2022
##   36   23    7   24   18

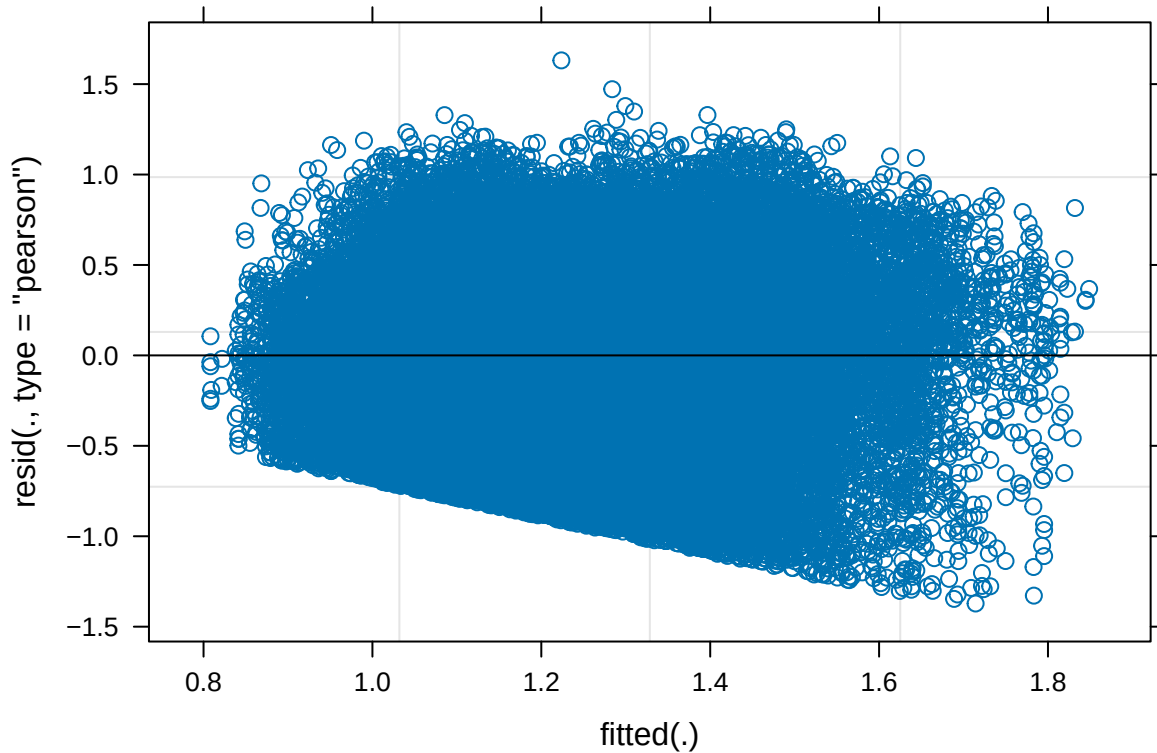
```

Question 1: Do knots stay longer in residence patches at night?

```

model_dur <- lmer(data=df, log_duration_min ~ tod_start*inout + (1|year) + (1|tag) + (1|tideID))
plot(model_dur)

```



```
#anova(model_dur, type="III")
summary(model_dur)
```

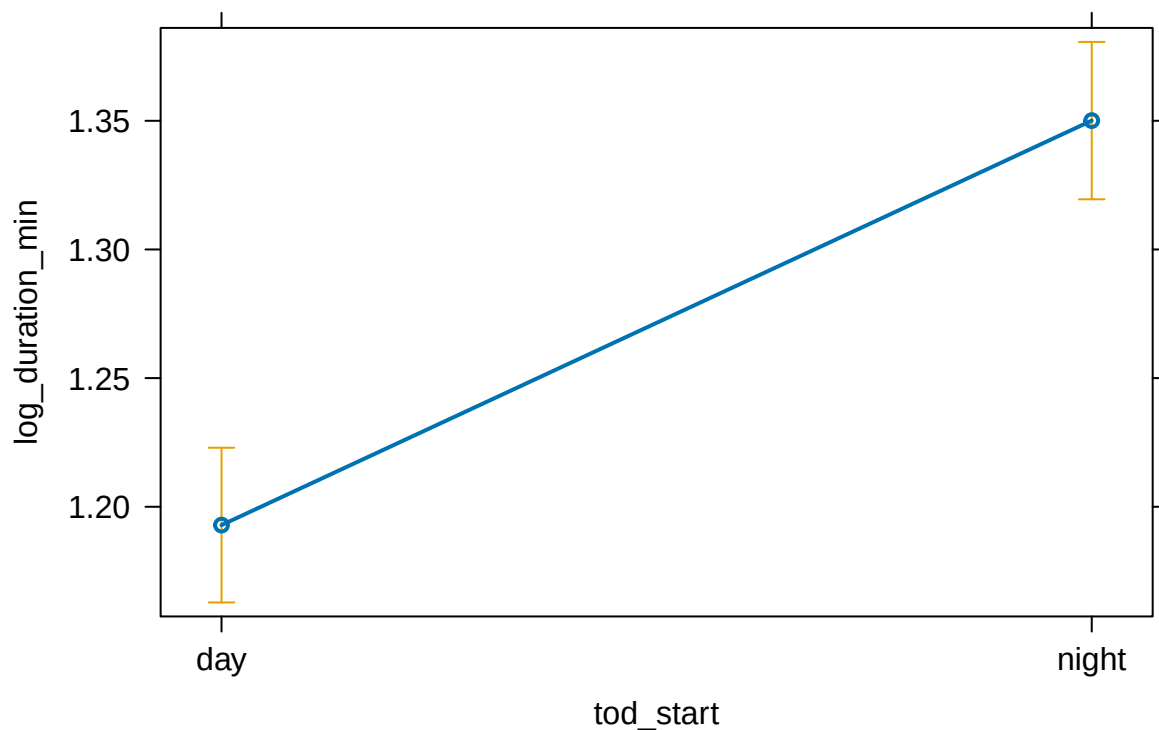
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log_duration_min ~ tod_start * inout + (1 | year) + (1 | tag) +
## (1 | tideID)
## Data: df
##
## REML criterion at convergence: 73548.8
##
## Scaled residuals:
##   Min      1Q  Median      3Q      Max
## -3.1032 -0.7199  0.0311  0.7287  3.6902
##
## Random effects:
##   Groups   Name                Variance Std.Dev.
##   tag      (Intercept)  0.0018057  0.04249
##   tideID   (Intercept)  0.0114596  0.10705
##   year     (Intercept)  0.0008307  0.02882
##   Residual                    0.1954371  0.44208
## Number of obs: 60276, groups: tag, 376; tideID, 236; year, 5
##
## Fixed effects:
##
##               Estimate Std. Error    df t value
```

```
## (Intercept)          1.106389      0.015440      4.373570      71.66
## tod_startnight      0.074838      0.006803 48580.886070      11.00
## inoutoutgoing      0.187572      0.004567 58242.160568      41.07
## tod_startnight:inoutoutgoing 0.178623      0.009433 50976.433728      18.94
##                               Pr(>|t|)
## (Intercept)          0.0000000697 ***
## tod_startnight      < 0.0000000000000002 ***
## inoutoutgoing      < 0.0000000000000002 ***
## tod_startnight:inoutoutgoing < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) td_str inttgn
## td_strtnight -0.135
## inoutoutgng  -0.117  0.274
## td_strtngh:   0.054 -0.563 -0.574
```

```
plot(effect("tod_start", model_dur))
```

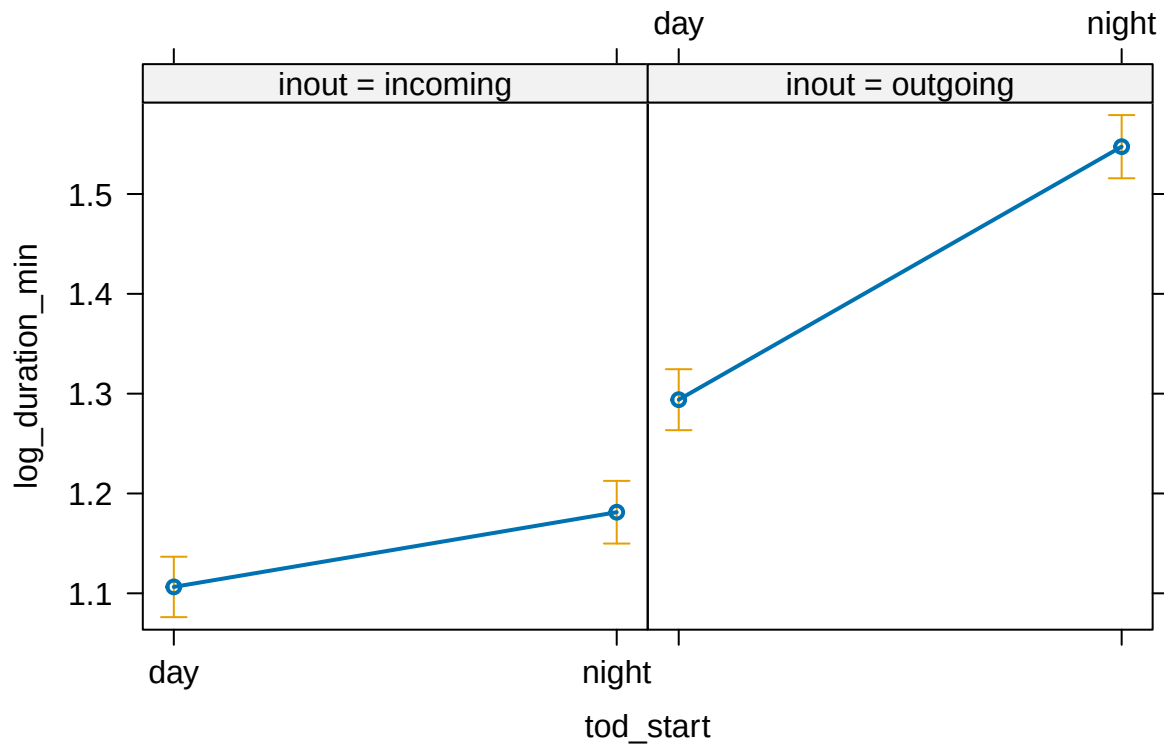
NOTE: tod_start is not a high-order term in the model

tod_start effect plot



```
plot(effect("tod_start*inout", model_dur))
```

tod_start*inout effect plot



```
duration_df <- df[,c("tod_start", "tag", "year", "tideID", "inout")]
duration_df$log_duration_min <- predict(model_dur, newdata=duration_df, type="response")
duration_df$duration_min <- 10^duration_df$log_duration_min
duration_df[duration_df$inout == "incoming",]$inout <- "flood"
duration_df[duration_df$inout == "outgoing",]$inout <- "ebb"

#flood means
mean(duration_df[duration_df$inout == "flood" & duration_df$tod_start=="day",]$duration_min)

## [1] 13.29858

sd(duration_df[duration_df$inout == "flood" & duration_df$tod_start=="day",]$duration_min)

## [1] 3.300644

mean(duration_df[duration_df$inout == "flood" & duration_df$tod_start=="night",]$duration_min)

## [1] 14.6325

sd(duration_df[duration_df$inout == "flood" & duration_df$tod_start=="night",]$duration_min)

## [1] 3.682526
```

```

#ebb means
mean(duration_df[duration_df$inout == "ebb" & duration_df$tod_start=="day",]$duration_min)

## [1] 21.77256

sd(duration_df[duration_df$inout == "ebb" & duration_df$tod_start=="day",]$duration_min)

## [1] 5.104927

mean(duration_df[duration_df$inout == "ebb" & duration_df$tod_start=="night",]$duration_min)

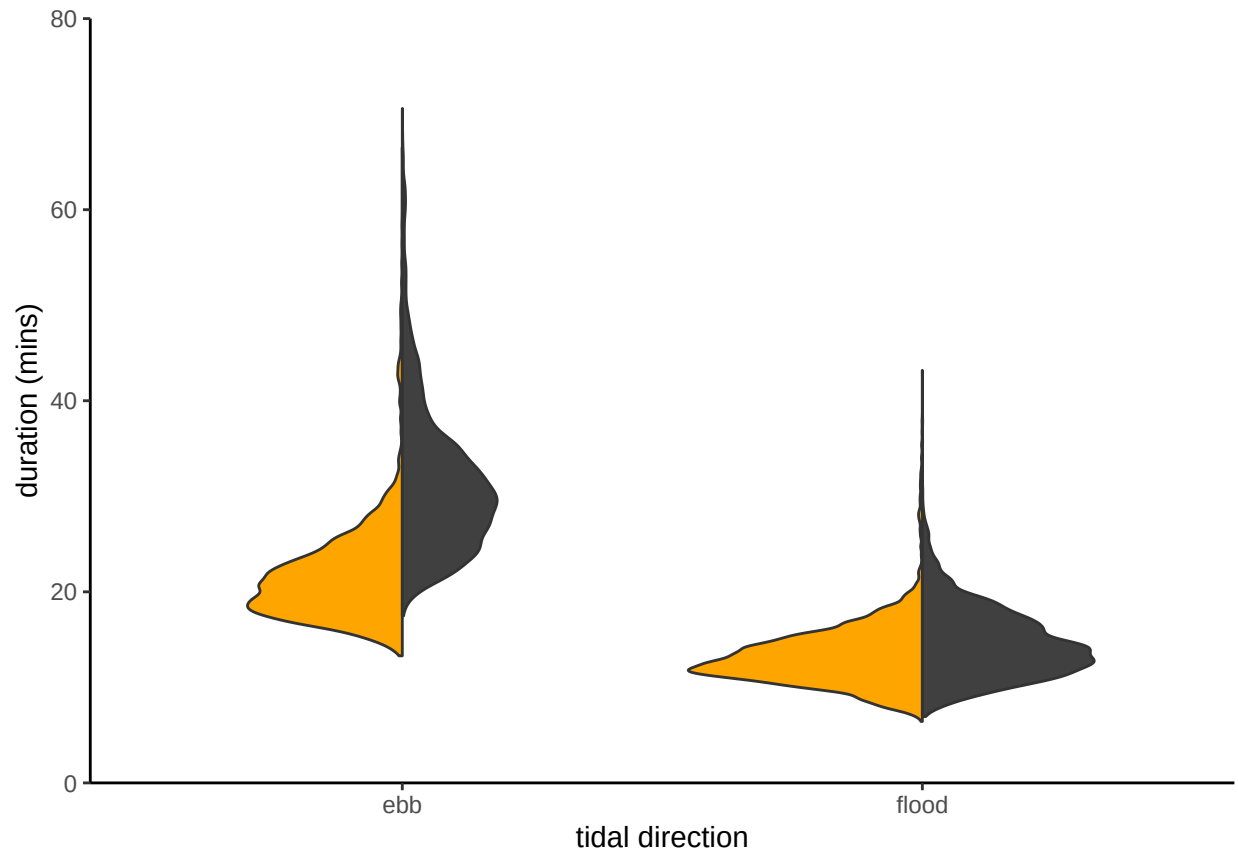
## [1] 31.38902

sd(duration_df[duration_df$inout == "ebb" & duration_df$tod_start=="night",]$duration_min)

## [1] 7.73373

p1 <- ggplot(data= duration_df)+
  geom_split_violin(aes(y = duration_min, x = inout, fill=tod_start))+
  scale_fill_manual(values=c("orange","grey25"))+
  scale_y_continuous(expand=c(0,0), limits = c(0,80), name = "duration (mins)")+
  scale_x_discrete(name = "tidal direction")+
  theme_classic()+
  theme(legend.position="none")
p1

```



Question 2: Do red knots stay closer to Griend at night

```
bird_tides$low_level_scaled <- scale(bird_tides$low_level, center = T, scale = F) # center subtracts the
scale_c <- attributes(bird_tides$low_level_scaled)$`scaled:center`

model_furth <- lmer(data=bird_tides, furthest_patch ~ tide_tod + low_level_scaled + (1|year) + (1|tag))
anova(model_furth, type="III")
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq   Mean Sq NumDF   DenDF  F value           Pr(>F)
## tide_tod        5974829   5974829      1 2186.0    5.5359      0.01872
## low_level_scaled 114750219 114750219      1 2187.4  106.3196 < 0.0000000000000002
##
## tide_tod          *
## low_level_scaled ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_furth)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: furthest_patch ~ tide_tod + low_level_scaled + (1 | year) + (1 |
##   tag)
##   Data: bird_tides
##
## REML criterion at convergence: 38360.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.7717 -0.7309 -0.0208  0.7157  3.4372
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   tag      (Intercept) 359582   599.7
##   year     (Intercept) 19534    139.8
##   Residual                1079296 1038.9
## Number of obs: 2281, groups: tag, 108; year, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value      Pr(>|t|)
## (Intercept)   2456.1685    92.2007    1.8648  26.639      0.00201
## tide_todnight -108.0233    45.9119  2185.9575  -2.353      0.01872
## low_level_scaled -7.3077     0.7087  2187.3784 -10.311 < 0.0000000000000002
##
## (Intercept)    **
## tide_todnight   *
## low_level_scaled ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```



```
## Correlation of Fixed Effects:
##          (Intr) td_tdn
## tide_tdngh  -0.189
## lw_lvl_scl  -0.009  0.019
```

```
coef(model_furth)
```

```
## $tag
##      (Intercept) tide_todnight low_level_scaled
## 0468    2470.226      -108.0233      -7.307682
## 0473    2623.409      -108.0233      -7.307682
## 0476    2489.115      -108.0233      -7.307682
## 0478    2723.236      -108.0233      -7.307682
## 0480    3228.443      -108.0233      -7.307682
## 0481    2770.332      -108.0233      -7.307682
## 0482    2346.688      -108.0233      -7.307682
## 0489    2711.734      -108.0233      -7.307682
## 0490    3424.825      -108.0233      -7.307682
## 0491    2175.995      -108.0233      -7.307682
## 0492    2121.185      -108.0233      -7.307682
## 0496    2148.264      -108.0233      -7.307682
## 0499    2738.660      -108.0233      -7.307682
## 0439    2558.373      -108.0233      -7.307682
## 0504    1805.686      -108.0233      -7.307682
## 0505    2306.608      -108.0233      -7.307682
## 0509    1656.694      -108.0233      -7.307682
## 0513    1979.021      -108.0233      -7.307682
## 0520    3131.558      -108.0233      -7.307682
## 0526    2214.106      -108.0233      -7.307682
## 0534    2081.016      -108.0233      -7.307682
## 0541    3377.352      -108.0233      -7.307682
## 0542    1958.850      -108.0233      -7.307682
## 0543    3478.089      -108.0233      -7.307682
## 0551    2108.334      -108.0233      -7.307682
## 0554    2065.862      -108.0233      -7.307682
## 0560    2547.320      -108.0233      -7.307682
## 0564    1849.643      -108.0233      -7.307682
## 0565    2073.247      -108.0233      -7.307682
## 0566    1985.607      -108.0233      -7.307682
## 0567    2088.329      -108.0233      -7.307682
## 0574    1794.744      -108.0233      -7.307682
## 0576    1358.034      -108.0233      -7.307682
## 0579    2461.603      -108.0233      -7.307682
## 0586    2979.107      -108.0233      -7.307682
## 0506    2297.027      -108.0233      -7.307682
## 0789    2659.123      -108.0233      -7.307682
## 0797    3615.699      -108.0233      -7.307682
## 0799    1877.537      -108.0233      -7.307682
## 0806    2421.051      -108.0233      -7.307682
## 0832    3483.947      -108.0233      -7.307682
## 0833    3697.429      -108.0233      -7.307682
## 0793    2562.416      -108.0233      -7.307682
## 0800    2820.543      -108.0233      -7.307682
## 0804    2355.269      -108.0233      -7.307682
```

## 0807	3173.329	-108.0233	-7.307682
## 0815	1655.243	-108.0233	-7.307682
## 0828	2230.573	-108.0233	-7.307682
## 0835	1690.536	-108.0233	-7.307682
## 0843	2718.068	-108.0233	-7.307682
## 0844	1071.502	-108.0233	-7.307682
## 0849	2660.469	-108.0233	-7.307682
## 0850	1455.492	-108.0233	-7.307682
## 0854	3446.659	-108.0233	-7.307682
## 0859	1381.975	-108.0233	-7.307682
## 0814	2095.010	-108.0233	-7.307682
## 0816	2968.252	-108.0233	-7.307682
## 0860	3011.619	-108.0233	-7.307682
## 0916	1210.255	-108.0233	-7.307682
## 2037	2476.625	-108.0233	-7.307682
## 2117	2673.760	-108.0233	-7.307682
## 2133	2469.603	-108.0233	-7.307682
## 2141	2833.469	-108.0233	-7.307682
## 2167	3027.969	-108.0233	-7.307682
## 2020	3219.479	-108.0233	-7.307682
## 2110	3145.297	-108.0233	-7.307682
## 2342	2575.359	-108.0233	-7.307682
## 2348	3173.721	-108.0233	-7.307682
## 2384	2635.830	-108.0233	-7.307682
## 2385	2666.200	-108.0233	-7.307682
## 2389	2824.281	-108.0233	-7.307682
## 2390	2180.470	-108.0233	-7.307682
## 2392	2033.548	-108.0233	-7.307682
## 2395	2282.893	-108.0233	-7.307682
## 2396	2695.780	-108.0233	-7.307682
## 2398	2734.244	-108.0233	-7.307682
## 2410	2532.476	-108.0233	-7.307682
## 2440	2362.288	-108.0233	-7.307682
## 2449	2891.357	-108.0233	-7.307682
## 2452	1903.322	-108.0233	-7.307682
## 2454	2361.680	-108.0233	-7.307682
## 2455	2675.635	-108.0233	-7.307682
## 2462	2711.983	-108.0233	-7.307682
## 2465	2060.394	-108.0233	-7.307682
## 2482	2532.984	-108.0233	-7.307682
## 2490	1930.578	-108.0233	-7.307682
## 2493	2296.633	-108.0233	-7.307682
## 2494	2961.778	-108.0233	-7.307682
## 2502	1641.530	-108.0233	-7.307682
## 2476	2126.381	-108.0233	-7.307682
## 2701	2171.248	-108.0233	-7.307682
## 2714	2316.725	-108.0233	-7.307682
## 2715	2579.254	-108.0233	-7.307682
## 2716	3435.750	-108.0233	-7.307682
## 2718	3131.505	-108.0233	-7.307682
## 2721	2268.469	-108.0233	-7.307682
## 2722	2244.249	-108.0233	-7.307682
## 2724	3487.981	-108.0233	-7.307682
## 2709	2495.679	-108.0233	-7.307682

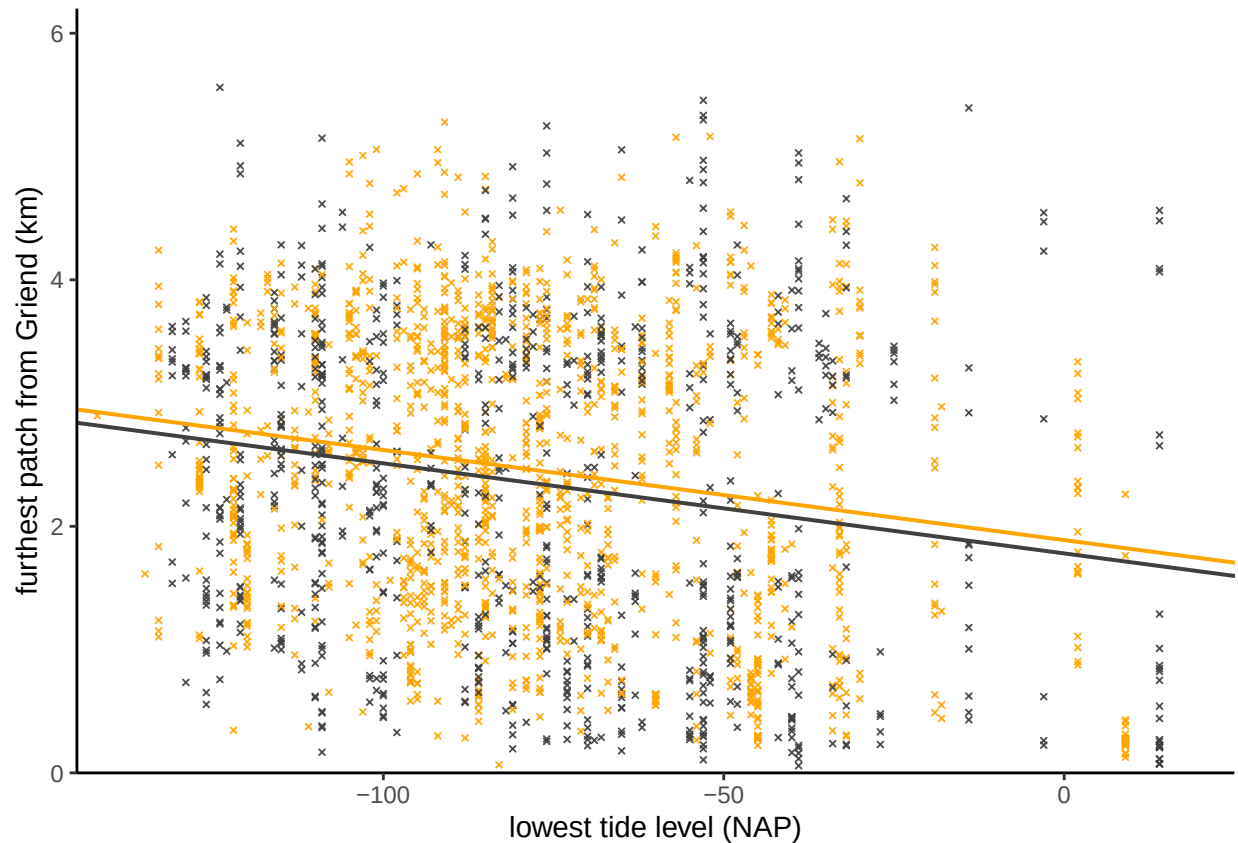
```
## 2742      2842.363      -108.0233      -7.307682
## 2748      2382.874      -108.0233      -7.307682
## 2758      1823.555      -108.0233      -7.307682
## 2788      1733.904      -108.0233      -7.307682
## 2800      2479.861      -108.0233      -7.307682
## 2815      1708.420      -108.0233      -7.307682
## 2833      2933.554      -108.0233      -7.307682
## 2840      2174.127      -108.0233      -7.307682
## 2835      2028.818      -108.0233      -7.307682
##
## $year
##      (Intercept) tide_todnight low_level_scaled
## 2018      2331.561      -108.0233      -7.307682
## 2019      2443.680      -108.0233      -7.307682
## 2020      2600.294      -108.0233      -7.307682
## 2021      2447.656      -108.0233      -7.307682
## 2022      2457.652      -108.0233      -7.307682
##
## attr(,"class")
## [1] "coef.mer"
```

```
intercept_day <- mean(c(coef(model_furth)$tag[,1],coef(model_furth)$year[,1])) - scale_c*coef(model_furth)$tag[1,"tide_todnight"]
intercept_night <- intercept_day + coef(model_furth)$tag[1,"tide_todnight"]
confint_day <- quantile(c(coef(model_furth)$tag[,1],coef(model_furth)$year[,1]), c(0.025, 0.975))

slope <- coef(model_furth)$tag[1,"low_level_scaled"]

# Predict
furth_df <- unique(as.data.frame(bird_tides[,c("tide_tod", "tag", "year", "tideID","low_level_scaled")]))
furth_df$furthest_patch <- predict(model_furth, newdata=furth_df,type="response")
furth_df$low_level <- furth_df$low_level_scaled + mean(bird_tides$low_level)

p2 <- ggplot(data = bird_tides, aes(x=low_level, y = furthest_patch/1000, col = tide_tod))+
  geom_point(alpha = 0.9, pch = 4, size=0.7)+
  geom_abline(intercept = intercept_day/1000, slope = slope/1000, col = "orange", linewidth = 0.7)+
  geom_abline(intercept = intercept_night/1000, slope = slope/1000, col = "grey25", linewidth = 0.7)+
  scale_fill_manual(values=c("orange","grey25"))+
  scale_color_manual(values=c("orange","grey25"))+
  scale_y_continuous(expand=c(0,0), limits = c(0,6.2), name = "furthest patch from Griend (km)")+
  scale_x_continuous(expand=c(0,0), limits = c(-145,25), name = "lowest tide level (NAP)")+
  theme_classic()+
  theme(legend.position="none")
p2
```



Question 3: Do knots use smaller areas at night?

```
b_tides <- st_read("C:/Users/BESCBEAR/OneDrive - Liverpool John Moores University/Synced files/Projects/
mutate(bird_tides = paste(tideID, id, sep="_"))
```

```
## Reading layer '2018-2022_bird_tides_multipolygons_min5tides' from data source
## 'C:\Users\BESCBEAR\OneDrive - Liverpool John Moores University\Synched files\Projects\1_NIOZ project
## using driver 'GPKG'
## Simple feature collection with 2306 features and 7 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: 645999.5 ymin: 5900627 xmax: 656482.5 ymax: 5907160
## Projected CRS: WGS 84 / UTM zone 31N
```

```
area_ptide <- b_tides %>%
  arrange(id) %>%
  mutate(area_ptide = (area_km2/duration_tracked)*6*60) # area covered in a 6h low tide period.

area_ptide$geom <- NULL

model_area <- lmer(data=area_ptide, area_ptide ~ tide_tod + (1|year) + (1|id))
anova(model_area, type="III")
```

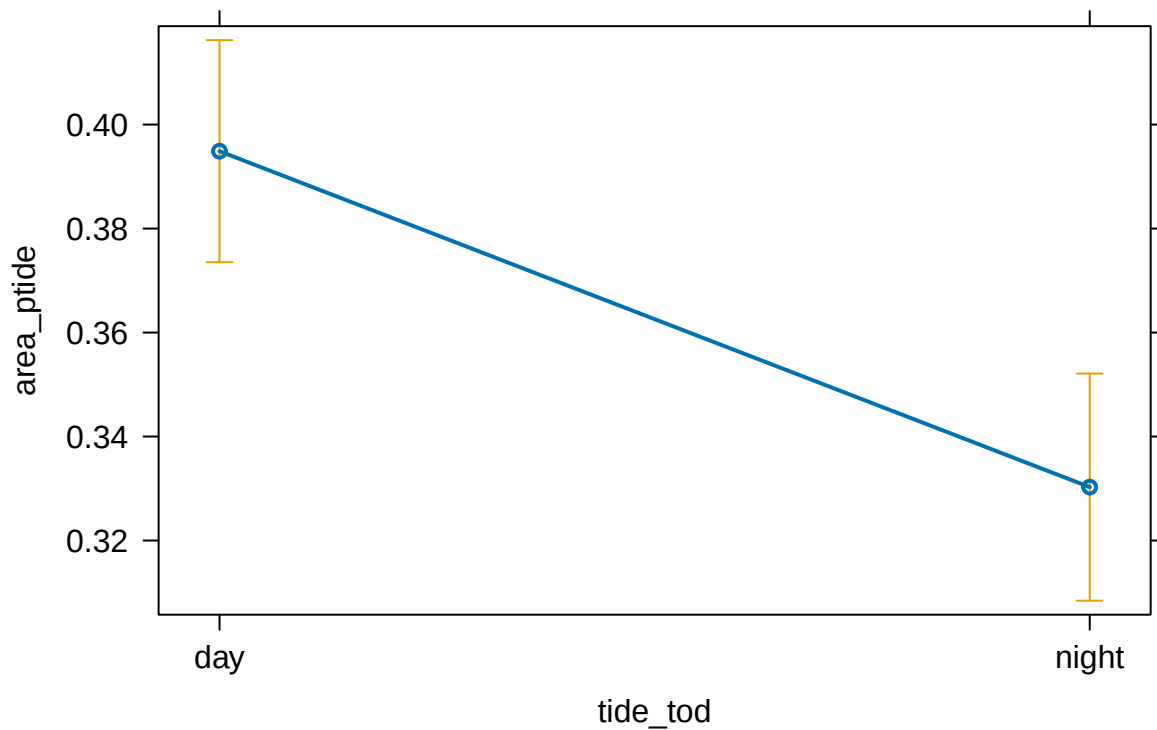
```
## Type III Analysis of Variance Table with Satterthwaite's method
##      Sum Sq Mean Sq NumDF  DenDF F value      Pr(>F)
## tide_tod    2.17    2.17     1 2222.4  169.76 < 0.00000000000000022 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_area)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: area_ptide ~ tide_tod + (1 | year) + (1 | id)
## Data: area_ptide
##
## REML criterion at convergence: -3347.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.9664 -0.6220 -0.0242  0.6448  4.6356
##
## Random effects:
## Groups Name Variance Std.Dev.
## id      (Intercept) 0.0016371 0.04046
## year    (Intercept) 0.0004441 0.02107
## Residual              0.0127828 0.11306
## Number of obs: 2306, groups: id, 108; year, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value      Pr(>|t|)
## (Intercept)    0.394877   0.010888   4.566029   36.27    0.000000852
## tide_todnight -0.064600   0.004958 2222.361697  -13.03 < 0.0000000000000002
##
## (Intercept) ***
## tide_todnight ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## tide_tdnghnt -0.177
```

```
plot(effect("tide_tod", model_area))
```

tide_tod effect plot



```
area_df <- area_ptide[,c("tide_tod", "id", "year")]
area_df$area_pred <- predict(model_area, newdata=area_df, type="response")

#mean predicted area for night
mean(area_df[area_df$tide_tod=="night",]$area_pred)
```

```
## [1] 0.3304045
```

```
sd(area_df[area_df$tide_tod=="night",]$area_pred)
```

```
## [1] 0.03974139
```

```
#mean predicted area during the day
mean(area_df[area_df$tide_tod=="day",]$area_pred)
```

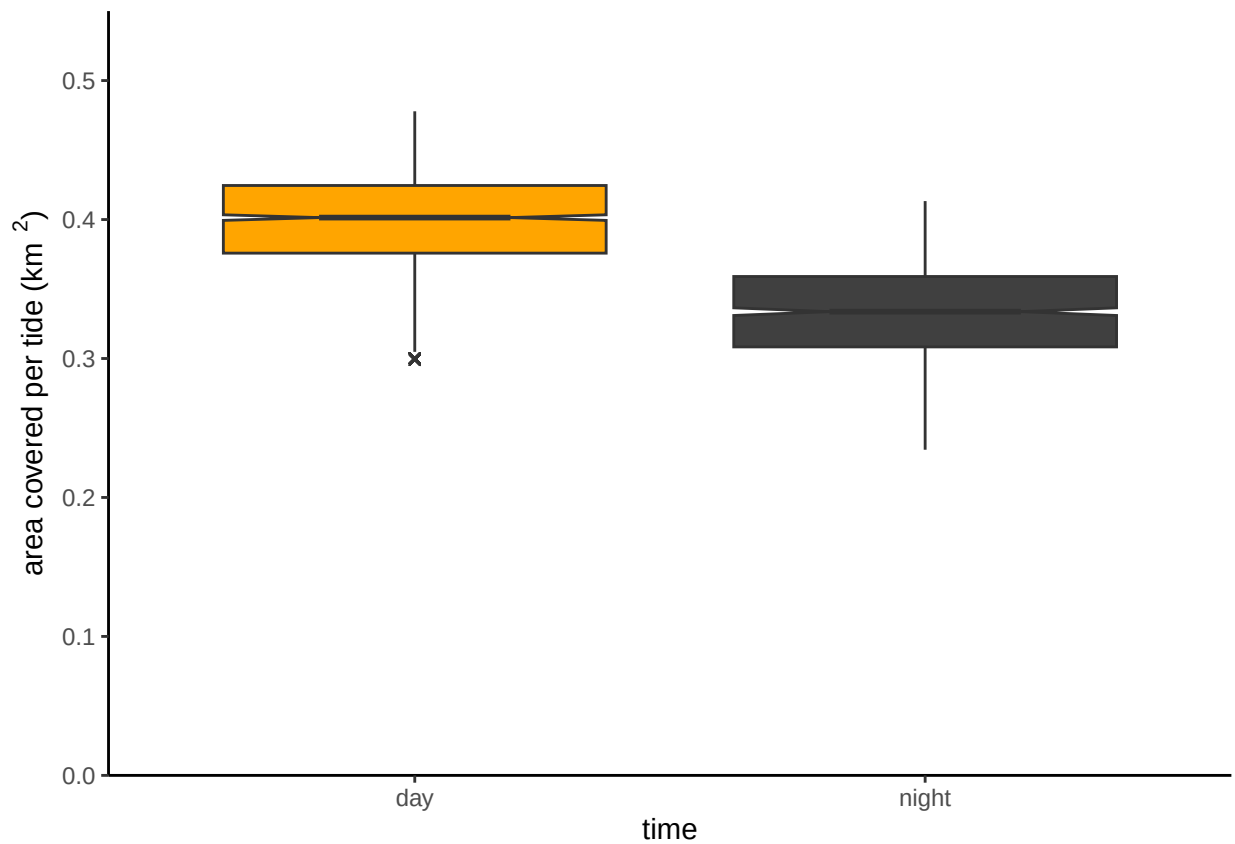
```
## [1] 0.3984496
```

```
sd(area_df[area_df$tide_tod=="day",]$area_pred)
```

```
## [1] 0.03954141
```

```
#predicted data
p3 <- ggplot(data = area_df, aes(x=tide_tod, y = area_pred, fill = tide_tod))+
  #geom_point(data = area_ptide, aes(x=tide_tod, y = area_ptide, fill = tide_tod), position = position_jitter)+
  geom_boxplot(outlier.shape= 4, notch=T)+
  scale_fill_manual(values = c("orange","grey25"))+
  scale_x_discrete(name = "time")+
  scale_y_continuous(limits = c(0,0.55), expand = c(0,0), name = expression(paste("area covered per tide (km2)")))+
  theme_classic()+
  theme(legend.position="none")
```

p3



Question 4: Do day and night areas overlap?

merge polygons for each bird (overlap) and calculate area used per tide

```
overlap <- st_read("C:/Users/BESCBEAR/OneDrive - Liverpool John Moores University/Synced files/Projects,
  group_by(year, id, tide_tod) %>%
  summarise(geom= st_union(geom),
    duration = sum(duration_tracked),
    num_tides = n()) %>%
  ungroup() %>%
  arrange(id)
```

```
## Reading layer '2018-2022_bird_tides_multipolygons_min5tides' from data source
## 'C:\Users\BESCBEAR\OneDrive - Liverpool John Moores University\Synced files\Projects\1_NIOZ project'
## using driver 'GPKG'
## Simple feature collection with 2306 features and 7 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: 645999.5 ymin: 5900627 xmax: 656482.5 ymax: 5907160
## Projected CRS: WGS 84 / UTM zone 31N

## 'summarise()' has grouped output by 'year', 'id'. You can override using the
## '.groups' argument.
```

```
setDT(overlap) #pivot wider no longer working with geom/sf

overlap<- as.data.frame(dcast(overlap, year+id ~ tide_tod, value.var = c("geom", "num_tides"))) %>%
  #tidyr::pivot_wider(names_from = "tide_tod", values_from = c("duration", "num_tides", "geom")) %>%
  st_set_geometry("geom_day") %>%
  group_by(id) %>%
  mutate(intersection = st_union(st_intersection(geom_day, geom_night)),
         area_day = as.numeric(st_area(geom_day))/10^6,
         area_night = as.numeric(st_area(geom_night))/10^6,
         area_intersect = as.numeric(st_area(intersection))/10^6,
         pc_overlap_day = as.numeric((area_intersect/area_day)*100),
         pc_overlap_night = as.numeric((area_intersect/area_night)*100),
         geom_day = NULL,
         geom_night = NULL,
         intersection = NULL) %>%
  as.data.frame()

#mean total space use at night km2
mean(overlap$area_night)
```

```
## [1] 2.053526
```

```
sd(overlap$area_night)
```

```
## [1] 0.7581772
```

```
#mean number of tides at night per bird
mean(overlap$num_tides_night)
```

```
## [1] 7.814815
```

```
sd(overlap$num_tides_night)
```

```
## [1] 2.510346
```

```
#mean total space use in day km2
mean(overlap$area_day)
```

```
## [1] 4.483422
```



```
sd(overlap$area_day)
```

```
## [1] 1.783366
```

```
#mean number of tides in day per bird
```

```
mean(overlap$num_tides_day)
```

```
## [1] 13.53704
```

```
sd(overlap$num_tides_day)
```

```
## [1] 5.116267
```

```
#mean percentage overlap of day tide with night tide space
```

```
mean(overlap$pc_overlap_day)
```

```
## [1] 19.95742
```

```
sd(overlap$pc_overlap_day)
```

```
## [1] 8.957541
```

```
#mean percentage overlap of night tide with day tide space
```

```
mean(overlap$pc_overlap_night)
```

```
## [1] 40.95783
```

```
sd(overlap$pc_overlap_night)
```

```
## [1] 11.90778
```

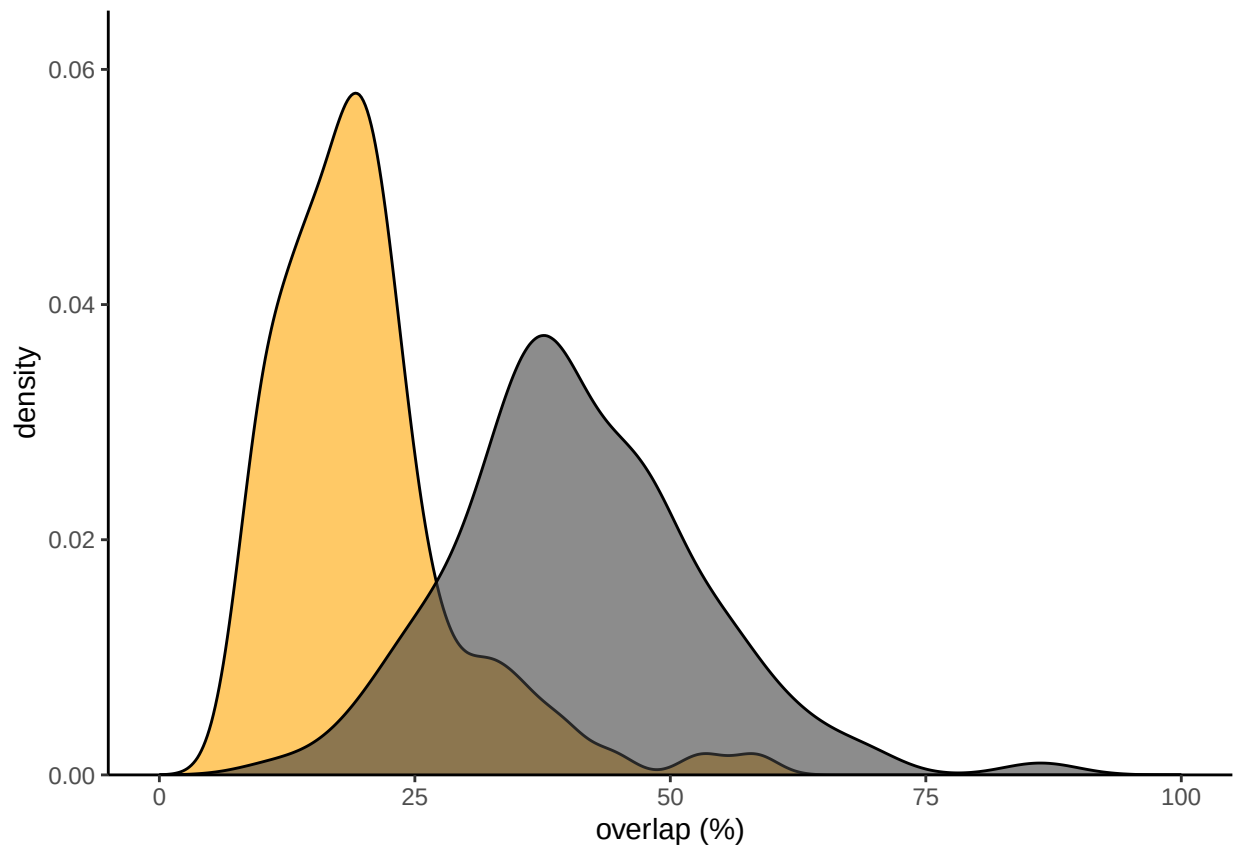
```
#plot for individuals
```

```
overlap_long <- overlap %>%
```

```
  tidyr::pivot_longer(cols = c("pc_overlap_night", "pc_overlap_day"),  
                      names_to = "time",  
                      values_to = "nonoverlap")
```

```
p4 <- ggplot(overlap_long, aes(x = nonoverlap, fill = time))+  
  geom_density(alpha = 0.6)+  
  scale_fill_manual(values=c("orange", "grey25"))+  
  scale_x_continuous(limits = c(0,100), name = "overlap (%)")+  
  scale_y_continuous(expand=c(0,0), limits = c(0,0.065))+  
  theme_classic()+  
  theme(legend.position="none")
```

```
p4
```



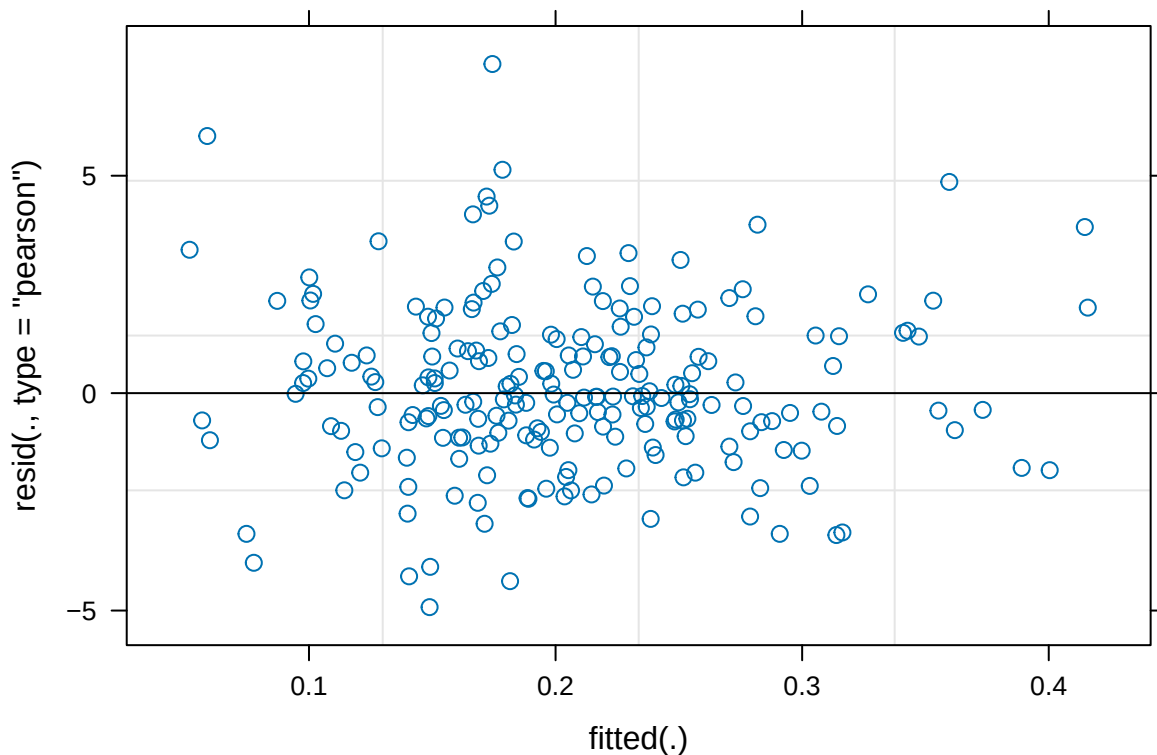
Question 5: Are birds more site faithful during the day or night?

```
summary_df <- read.csv("C:/Users/BESCBEAR/OneDrive - Liverpool John Moores University/Synced files/Proj
m_revisits1 <- glmer(cbind(visited2_cells, visited1_cells - visited2_cells) ~ tide_tod + offset(log(n_t
summary(m_revisits1)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: cbind(visited2_cells, visited1_cells - visited2_cells) ~ tide_tod +
## offset(log(n_tides)) + (1 | year) + (1 | id)
## Data: summary_df
##
##      AIC      BIC  logLik deviance df.resid
## 2540.8  2554.3 -1266.4  2532.8     212
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.9214 -0.9925 -0.0816  1.2921  7.5693
##
## Random effects:
## Groups Name          Variance Std.Dev.
## id      (Intercept) 0.122973 0.35068
```

```
## year (Intercept) 0.006669 0.08166
## Number of obs: 216, groups: id, 108; year, 5
##
## Fixed effects:
##           Estimate Std. Error z value      Pr(>|z|)
## (Intercept) -3.86821    0.05248  -73.71 <0.0000000000000002 ***
## tide_todnight 0.39421    0.01581   24.94 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## tide_tdnghgt -0.101
```

```
plot(m_revisits1)
```



```
#plot(effect("tide_tod", m_revisits1))

newdf1 <- data.frame(revisits = 1,
                     summary_df[,c("tide_tod", "id", "year")],
                     n_tides = mean(summary_df$n_tides))
newdf1$revisits_prob <- predict(m_revisits1, newdata=newdf1, type="response")

summary_df$revisits_pc <- summary_df$visited2_cells/summary_df$visited1_cells
```

```
p5 <- ggplot()+
  geom_boxplot(data= newdf1, aes(x= tide_tod, y = revisits_prob, fill = tide_tod), outlier.shape= 4,
  #geom_point(data= summary_df, aes(x= tide_tod, y = revisits_pc, fill = tide_tod), alpha=0.9, col="b
  scale_fill_manual(values=c("orange","grey25"))+
  scale_y_continuous(limits = c(0,0.5), expand = c(0,0), name = "revisit probability")+
  scale_x_discrete(name = "time")+
  theme_classic()+
  theme(legend.position="none")
```

p5

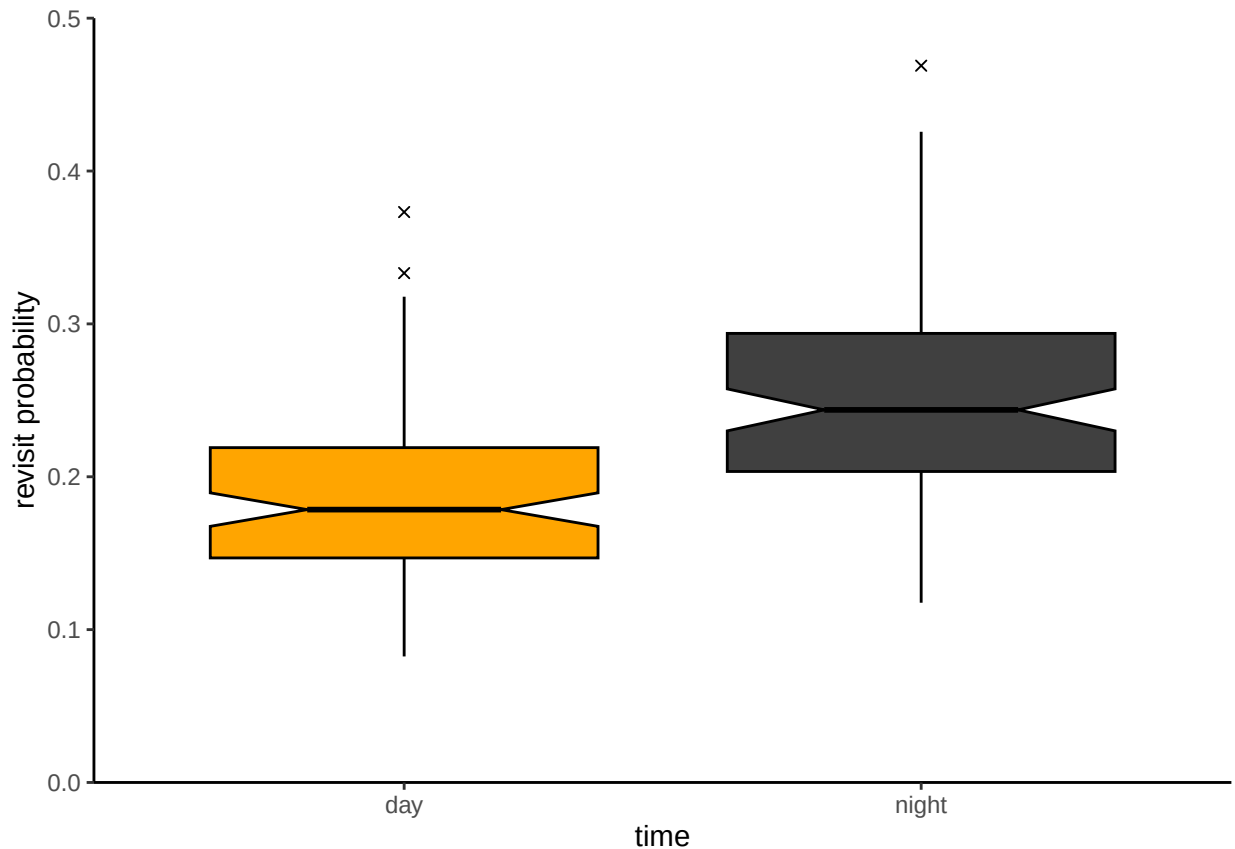


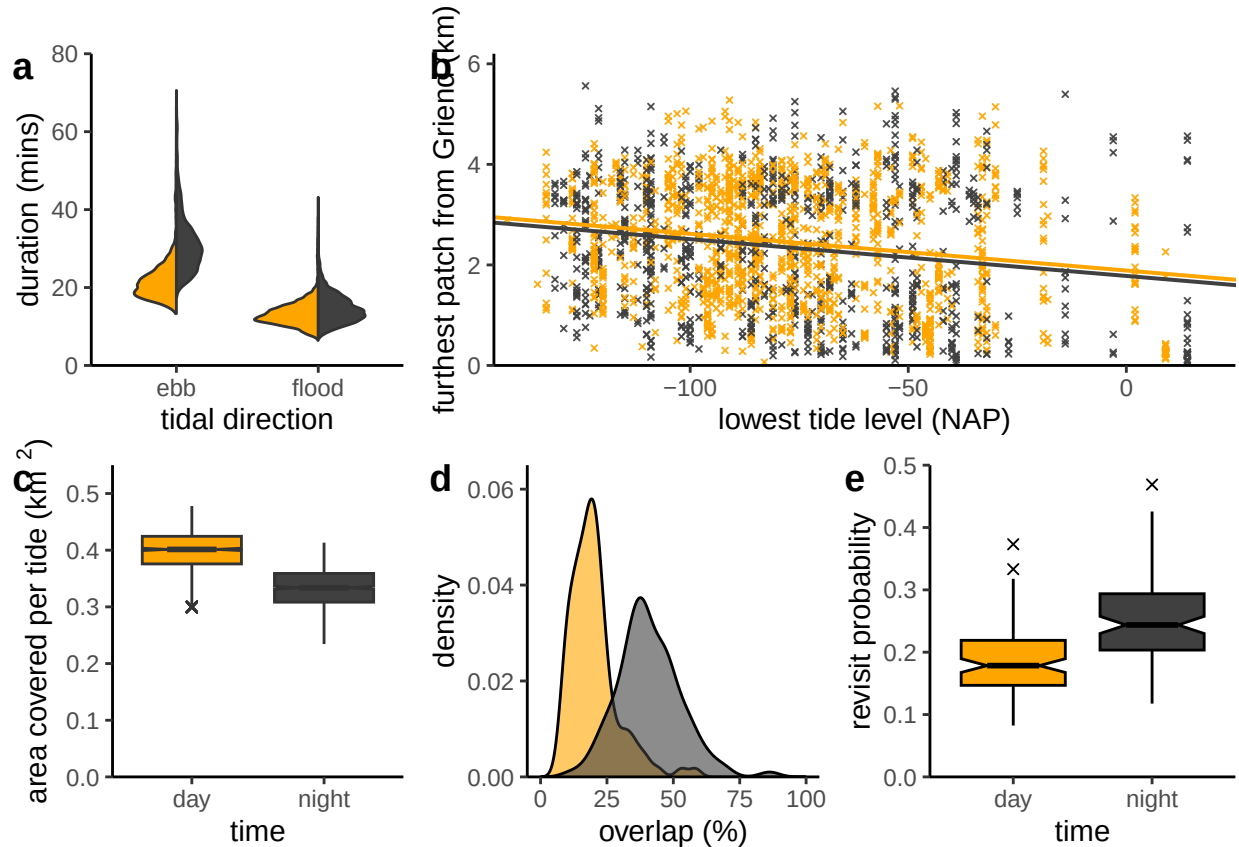
Fig. 2 merged plot

```
legend <- get_legend(p1 + theme(legend.position="top", legend.title = element_blank()))
```

```
## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use 'return_all = TRUE'.
```

```
r1 <- plot_grid(p1,p2, rel_widths = c(1,2), labels = c("a","b"))#, label_size=20)
r2<- plot_grid(p3,p4,p5, labels=c("c","d", "e"), nrow=1)#, label_size=20)
plot_all<- plot_grid(legend, r1, r2, nrow=3, rel_heights = c(0.1,1,1))
```

plot_all



#ggsave("F:\\OneDrive - Liverpool John Moores University\\Synced files\\Projects\\1_NIOZ projects\\2_Data")

For supplementary fig S1

different thresholds for site fidelity

```
m_revisits2 <- glmer(cbind(visited3_cells, visited1_cells - visited3_cells) ~ tide_tod + offset(log(n_tides)), data = summary_df)
m_revisits3 <- glmer(cbind(visited4_cells, visited1_cells - visited4_cells) ~ tide_tod + offset(log(n_tides)), data = summary_df)
m_revisits4 <- glmer(cbind(visited5_cells, visited1_cells - visited5_cells) ~ tide_tod + offset(log(n_tides)), data = summary_df)

newdf2 <- data.frame(revisits = 2,
  summary_df[,c("tide_tod", "id", "year")],
  n_tides = mean(summary_df$n_tides))
newdf2$revisits_prob <- predict(m_revisits2, newdata=newdf2, type="response")

newdf3 <- data.frame(revisits = 3,
  summary_df[,c("tide_tod", "id", "year")],
  n_tides = mean(summary_df$n_tides))
newdf3$revisits_prob <- predict(m_revisits3, newdata=newdf3, type="response")

newdf4 <- data.frame(revisits = 4,
```

```

summary_df[,c("tide_tod", "id", "year")],
n_tides = mean(summary_df$n_tides))
newdf4$revisits_prob <- predict(m_revisits4,newdata=newdf4,type="response")

newdf <- rbind(newdf1, newdf2, newdf3,newdf4)

realdf <- rbind(data.frame(revisits = 1, summary_df[,c("tide_tod", "id", "year")], revisits_pc = summar
data.frame(revisits = 2,summary_df[,c("tide_tod", "id", "year")], revisits_pc = summar
data.frame(revisits = 3, summary_df[,c("tide_tod", "id", "year")], revisits_pc = summar
data.frame(revisits = 4, summary_df[,c("tide_tod", "id", "year")], revisits_pc = summar

newdf2 <- merge(newdf, realdf, by = c("revisits", "tide_tod", "id", "year"))
newdf2$revisits = factor(newdf$revisits)

ggplot()+
  geom_boxplot(data= newdf2, aes(x= revisits, y = revisits_prob, fill = tide_tod), col="black", notch=TRUE)
  scale_fill_manual(values=c("orange","grey15"))+
  scale_color_manual(values=c("black", "black"))+
  facet_grid(cols = vars(year))+
  theme_bw()

```

